

NEXTorr[®] Z pumps

Technical specifications

M.H220T.0003.21 R3

Provisional

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Original English version.

The NEXTorr® product line has global Intellectual Property Rights coverage with patents already granted in the US (8,287,247), Europe (2,409,034), Japan (5,372,239), and China (102356236).

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Product description

NEXTorr® Z pumps are a family of vacuum combination pumps consisting of a non-evaporable getter (NEG) cartridge and a compact sputter-ion pump (ION), which are connected to the same ConFlat® (CF) flange.

The NEG element is made of sintered ZAO®1 (Ti-Zr-V-Al) porous disks, characterised by an extremely-low particle release; it is able to efficiently pump all chemically-reactive gases (H₂, H₂O, N₂, O₂, CO, CO₂), while the sputter-ion pump ensures a sufficient pumping for non-getterable species, such as CH₄ and noble gases.

NEXTorr Z pumps are designed and engineered to work in a variety of vacuum systems where stringent ultra-high vacuum and extremely-high vacuum (UHV-XHV) conditions should be met, such as particle accelerators, scanning electron microscopes, portable vacuum suitcases, and semiconductor inspection tools.

The following models are currently available:

- NEXTorr Z 100
- NEXTorr Z 200
- NEXTorr Z 300
- NEXTorr Z 500
- NEXTorr Z 1000
- NEXTorr Z 2000

The *Operating instructions* document provides all the information necessary to safely install and operate NEXTorr Z pumps.

Details about the activation parameters and the pumping performance of each NEXTorr Z model are given in the following pages.

NEXTorr Z 100

Pump dimensions

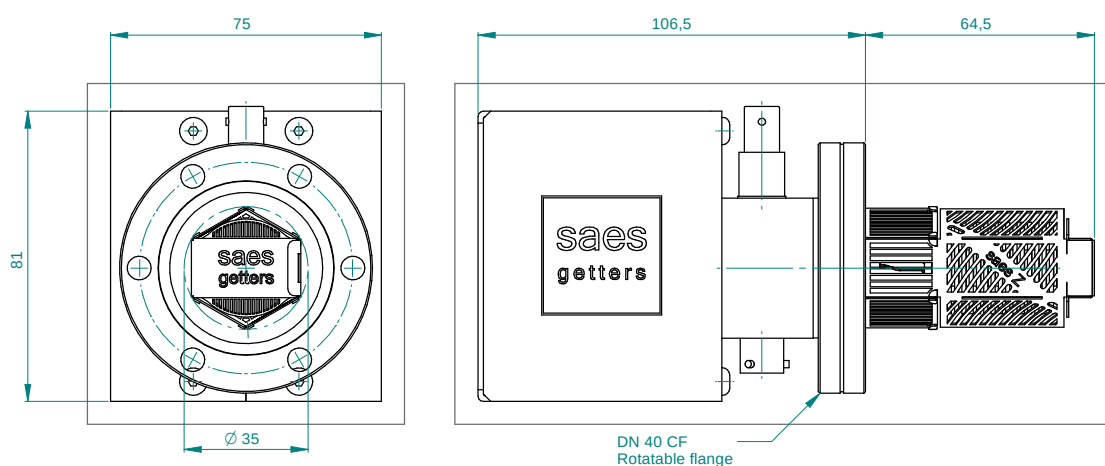


Figure 1 – 2D drawings of NEXTorr Z 100. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H200.0305.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	30
Getter surface [cm ²]	156
Flange size	CF35 (rotatable)
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	2.2
Maximum bake-out temperature [°C]	250*
ION element type	Diode

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

Activation and working parameters

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

Type of operation	Temperature	Electrical parameters
NEG activation	550 °C	45 W (9 V)*
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity † [Torr·l]
	Activated NEG‡	Saturated NEG§	
H ₂	150	6	600**
H ₂ O††	100	4	9
CO	75	5	0.6
N ₂	40	4	0.35
CH ₄	15		50,000 h at 10 ⁻⁶ Torr
Ar	6 (0.3)‡‡		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

* Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

† Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

‡ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

§ Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

** When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

†† Calculated values.

‡‡ Measured at 3·10⁻⁶ Torr. Unsaturated ION element (saturated ION element).

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive, *i.e.*, a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N_2 , air, or Ar) to vent the pump, avoiding the use of atmospheric air.

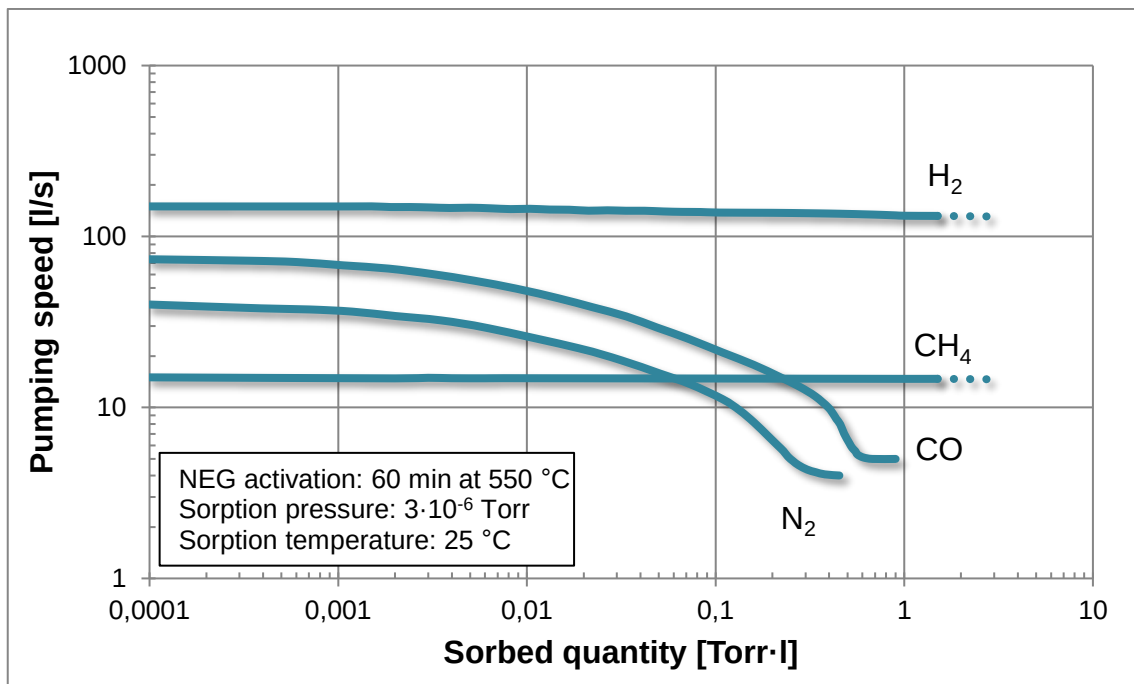


Figure 2 – Typical sorption curves of NEXTorr Z 100 (according to ASTM F798-97).

ION element pumping characteristics

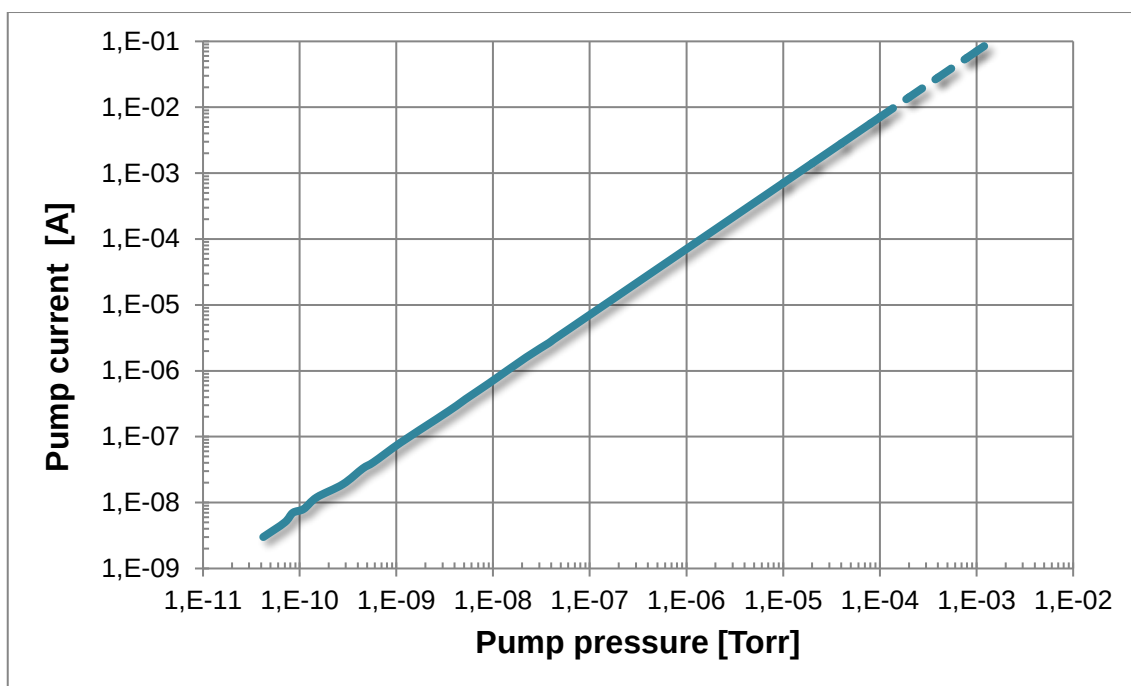


Figure 3 – ION current vs. pressure characteristic curve for N₂ of NEXTorr Z 100.

Ordering information

Product	Product description	Code
NEXTorr pump	NEXTorr Z 100	5H0211
NEG element power supply	NEG POWER MINI*	3B0110
NEG element output cable	NEG cable 3P5A 3 MT†	3B0598
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

NEXTorr Z 200

Pump dimensions

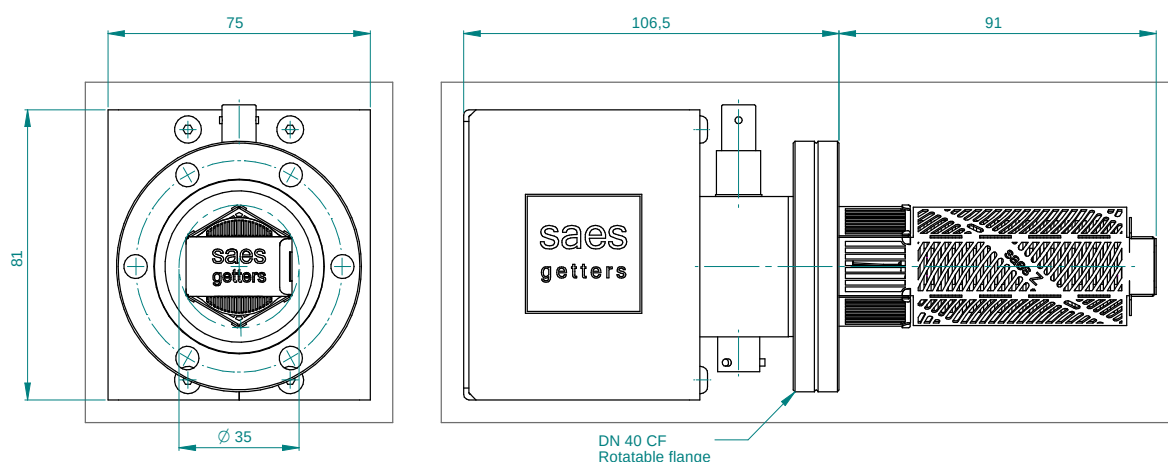


Figure 4 – 2D drawings of NEXTorr Z 200. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H200.0287.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	58
Getter surface [cm ²]	300
Flange size	CF35 (rotatable)
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	2.2
Maximum bake-out temperature [°C]	250*
ION element type	Diode

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

Activation and working parameters

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

Type of operation	Temperature	Electrical parameters
NEG activation	550 °C	57 W (12 V)*
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity † [Torr·l]
	Activated NEG‡	Saturated NEG§	
H ₂	290	6	1,160**
H ₂ O††	180	4	16
CO	140	5	1.5
N ₂	80	4	0.7
CH ₄	15		50,000 h at 10^{-6} Torr
Ar	6 (0.3)‡‡		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

* Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

† Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

‡ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

§ Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

** When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

†† Calculated values.

‡‡ Measured at $3 \cdot 10^{-6}$ Torr. Unsaturated ION element (saturated ION element).

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive, *i.e.*, a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N_2 , air, or Ar) to vent the pump, avoiding the use of atmospheric air.

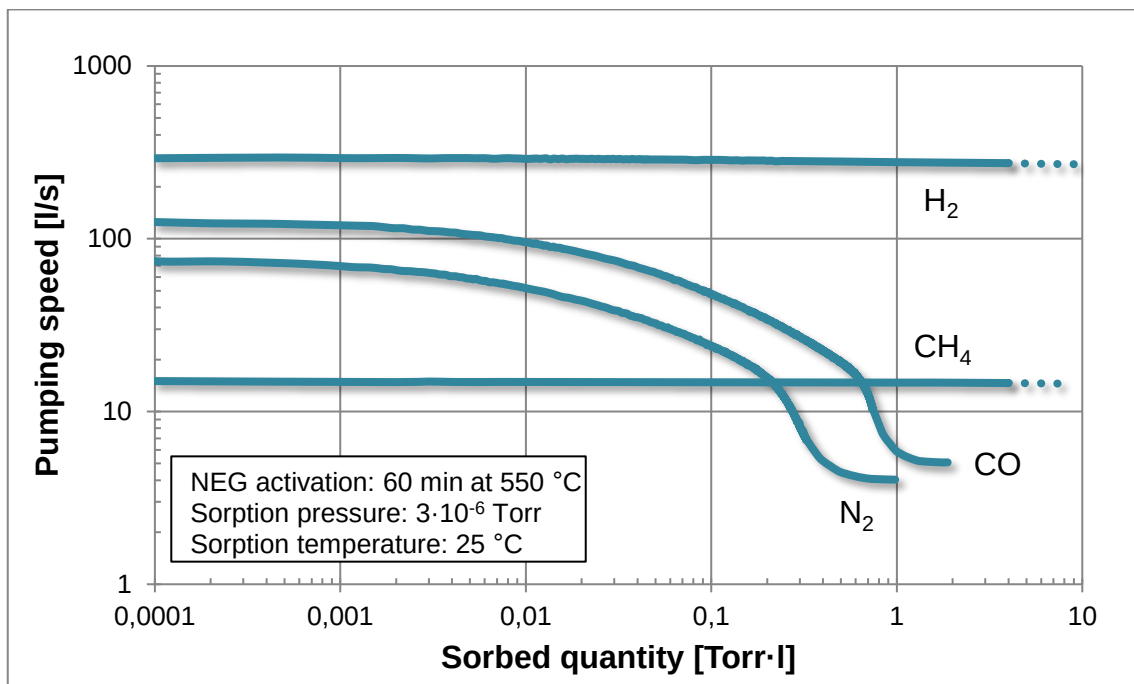


Figure 5 – Typical sorption curves of NEXTorr Z 200 (according to ASTM F798-97).

ION element pumping characteristics

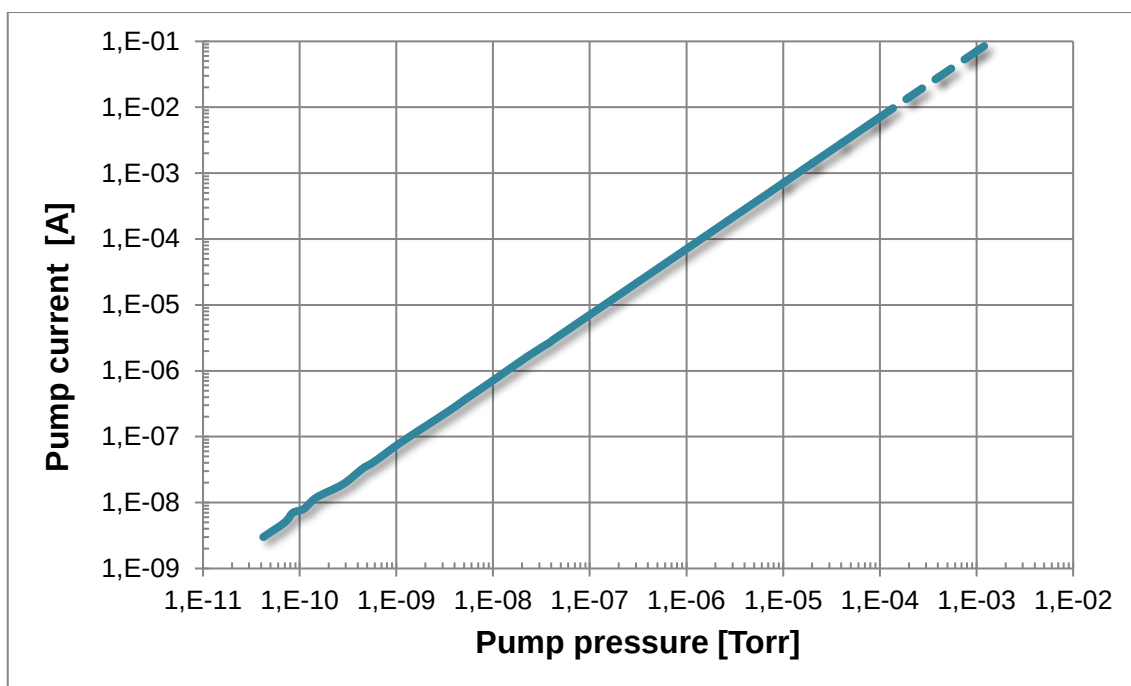


Figure 6 – ION current vs. pressure characteristic curve for N₂ of NEX Torr Z 200.

Ordering information

Product	Product description	Code
NEX Torr pump	NEX Torr Z 200	5H0199
NEG element power supply	NEG POWER MINI*	3B0110
NEG element output cable	NEG cable 3P5A 3 MT†	3B0598
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

NEXTorr Z 300

Pump dimensions

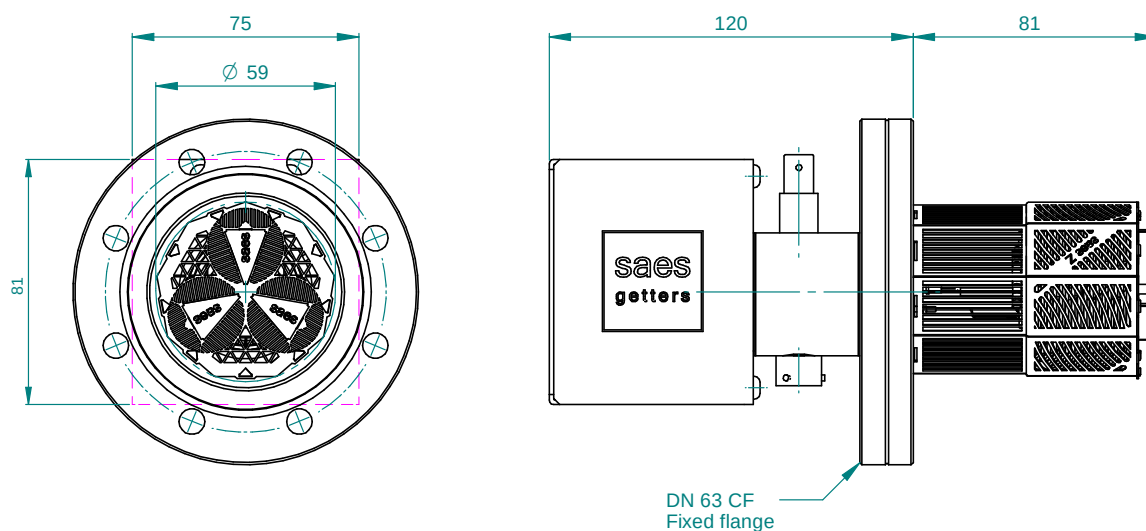


Figure 7 – 2D drawings of NEXTorr Z 300. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H200.0369.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	96
Getter surface [cm ²]	500
Flange size	CF63
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	3.1
Maximum bake-out temperature [°C]	250*
ION element type	Diode

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

Activation and working parameters

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

The pump is equipped with a type-K (Chromel-Alumel) thermocouple.

Type of operation	Temperature*	Electrical parameters
NEG activation	500 °C	111 W (24 V) [†]
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity ‡ [Torr·l]
	Activated NEG [§]	Saturated NEG ^{**}	
H ₂	400	6	1,920 ^{††}
H ₂ O ^{‡‡}	280	4	40
CO	200	5	2
N ₂	120	4	1.2
CH ₄	15		50,000 h at 10^{-6} Torr
Ar	6 (0.3) ^{§§}		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

* Thermocouple reading when the getter material is at 550 °C (activation).

† Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

‡ Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

§ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

** Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

†† When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

‡‡ Calculated values.

§§ Measured at $3 \cdot 10^{-6}$ Torr. Unsaturated ION element (saturated ION element).

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive, *i.e.*, a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N_2 , air, or Ar) to vent the pump, avoiding the use of atmospheric air.

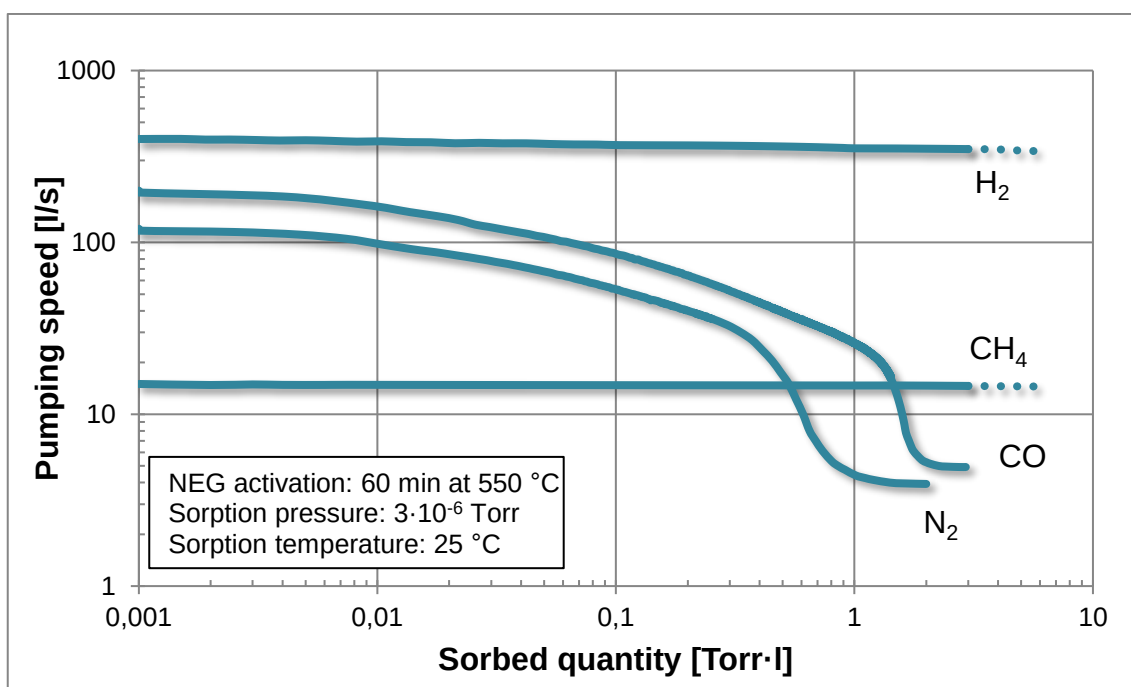


Figure 8 – Typical sorption curves of NEXTorr Z 300 (according to ASTM F798-97).

ION element pumping characteristics

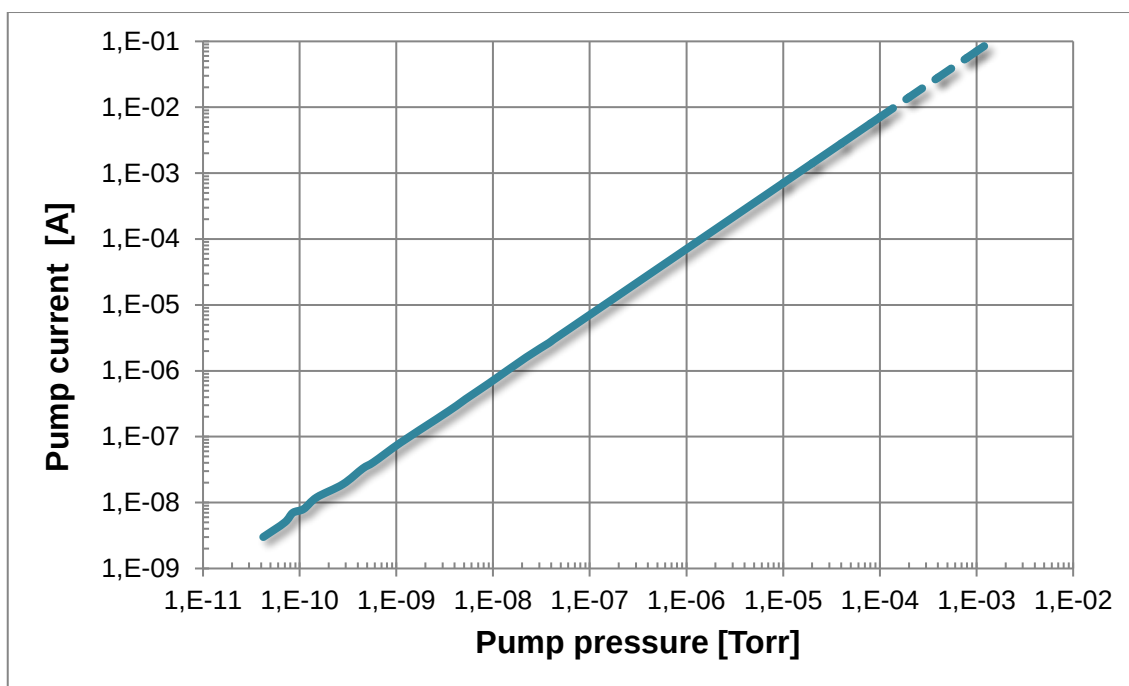


Figure 9 – ION current vs. pressure characteristic curve for N₂ of NEX Torr Z 300.

Ordering information

Product	Product description	Code
NEX Torr pump	NEX Torr Z 300	5H0201
NEG element power supply	NEG POWER MINI*	3B0110
NEG element output cable	NEG cable 6P5A 3 MT†	3B0854
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

NEXTorr Z 500

Pump dimensions

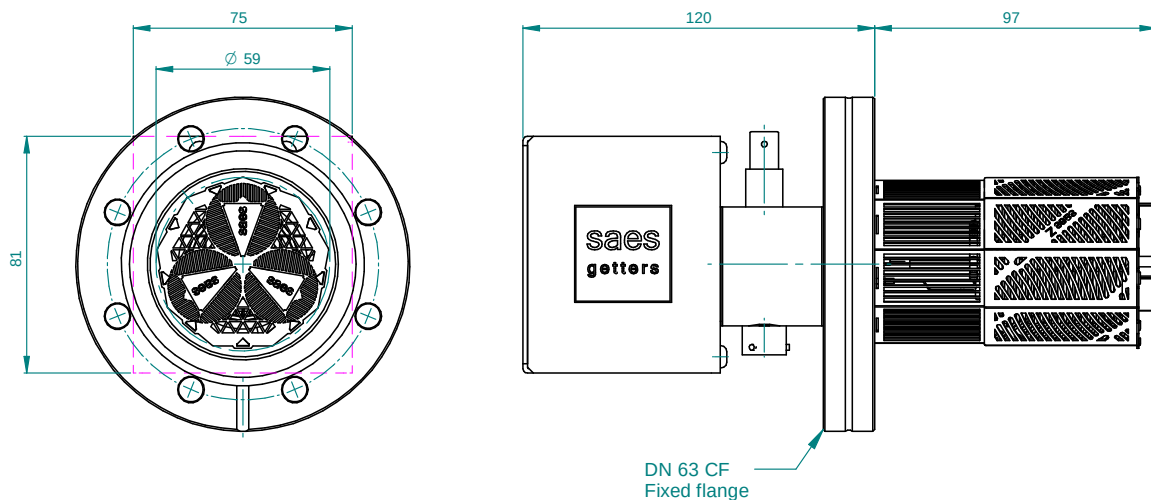


Figure 10 – 2D drawings of NEXTorr Z 500. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H220D.0003.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	144
Getter surface [cm ²]	750
Flange size	CF63
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	3.1
Maximum bake-out temperature [°C]	250*
ION element type	Diode

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

Activation and working parameters

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

The pump is equipped with a type-K (Chromel-Alumel) thermocouple.

Type of operation	Temperature*	Electrical parameters
NEG activation	500 °C	125 W (29 V) [†]
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity ‡ [Torr·l]
	Activated NEG [§]	Saturated NEG ^{**}	
H ₂	580	6	2,880 ^{††}
H ₂ O ^{‡‡}	440	4	60
CO	300	5	2.5
N ₂	180	4	1.5
CH ₄	15		50,000 h at 10^{-6} Torr
Ar	6 (0.3) ^{§§}		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

* Thermocouple reading when the getter material is at 550 °C (activation).

† Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

‡ Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

§ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

** Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

†† When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

‡‡ Calculated values.

§§ Measured at $3 \cdot 10^{-6}$ Torr. Unsaturated ION element (saturated ION element).

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive, *i.e.*, a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N_2 , air, or Ar) to vent the pump, avoiding the use of atmospheric air.

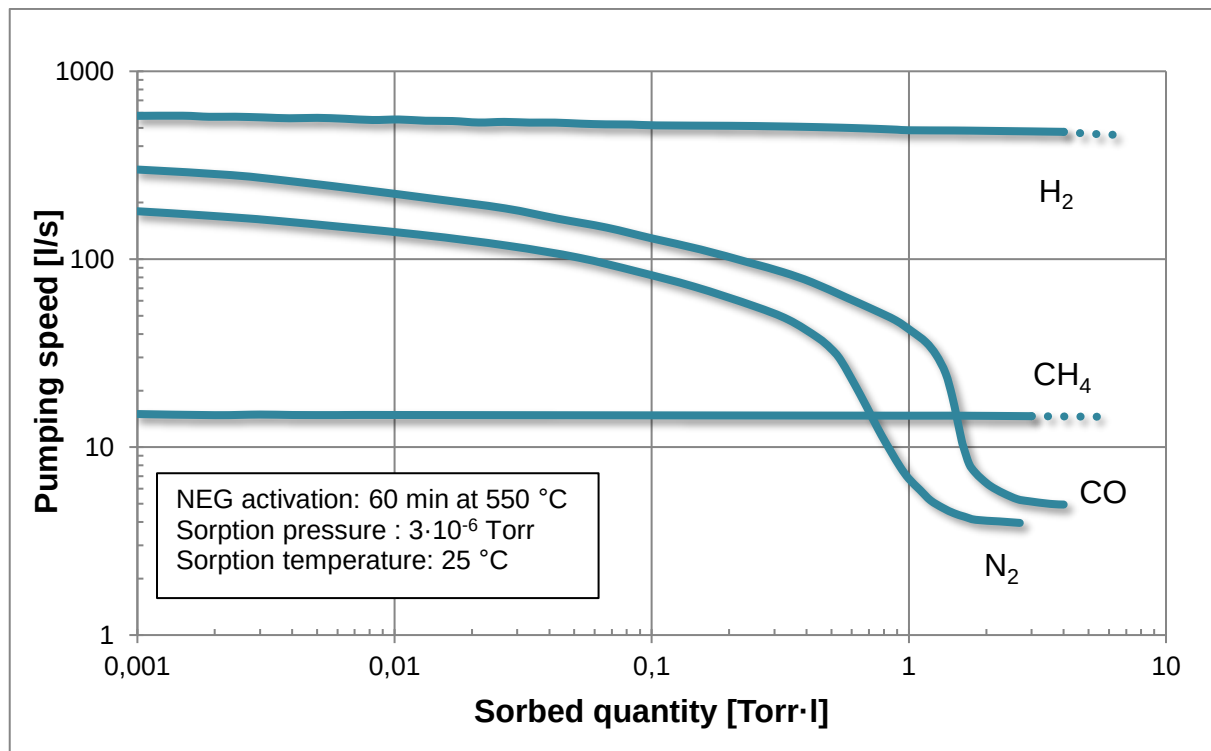


Figure 11 – Typical sorption curves of NEXTorr Z 500 (according to ASTM F798-97).

ION element pumping characteristics

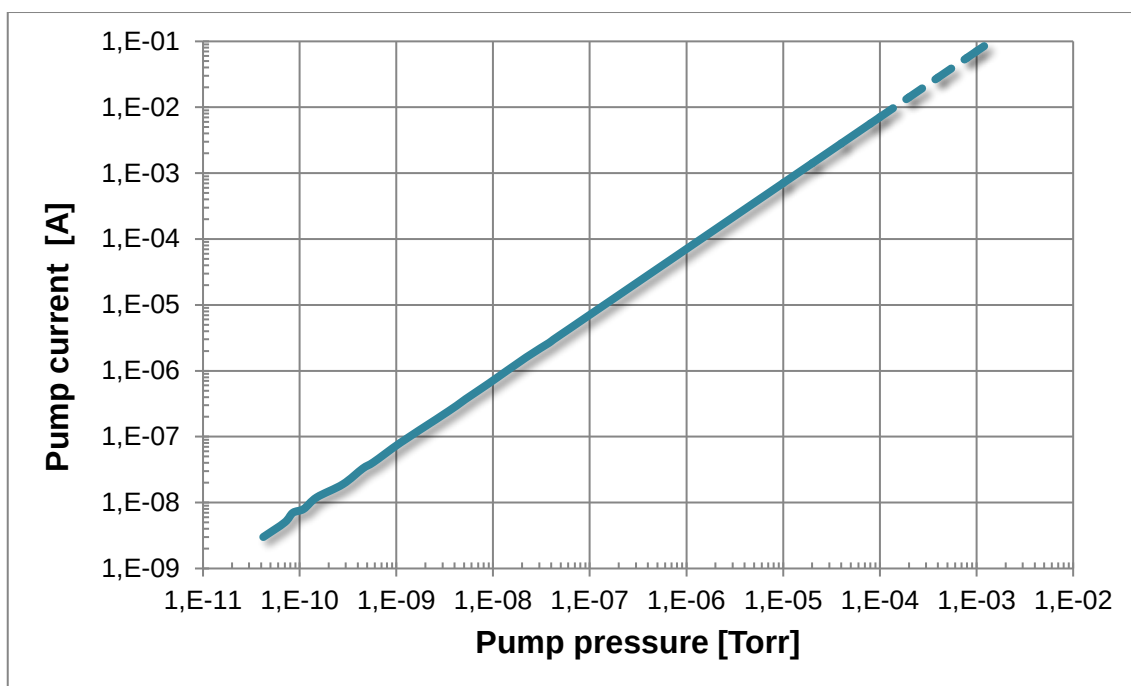


Figure 12 – ION current vs. pressure characteristic curve for N₂ of NEXTorr Z 500.

Ordering information

Product	Product description	Code
NEXTorr pump	NEXTorr Z 500	5H0234
NEG element power supply	NEG POWER MINI*	3B0110
NEG element output cable	NEG cable 6P5A 3 MT†	3B0854
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

NEXTorr Z 1000

Pump dimensions

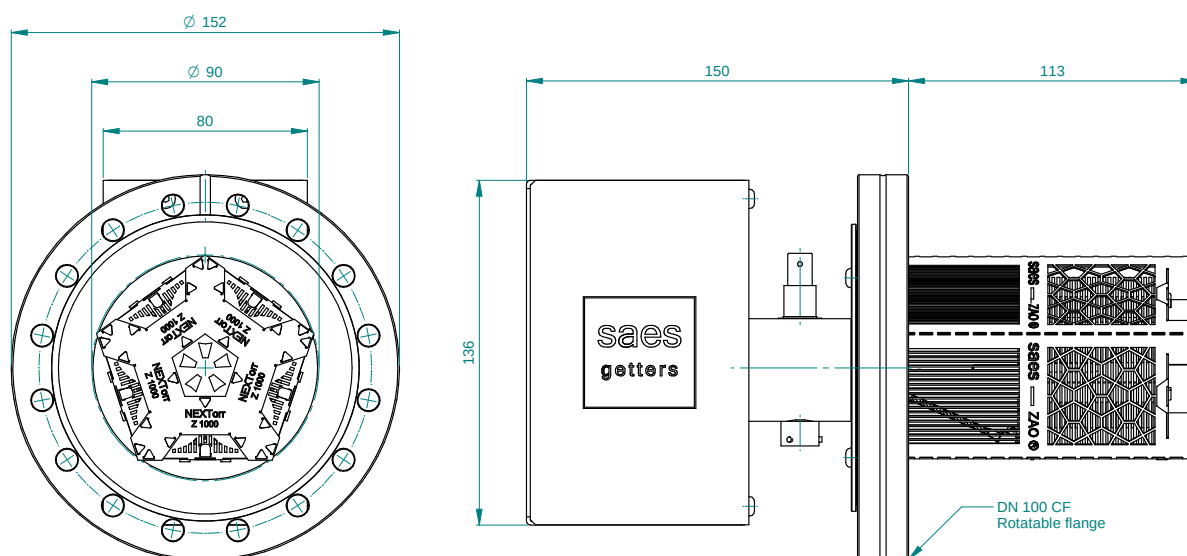


Figure 13 – 2D drawings of NEXTorr Z 1000. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H220.0002.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	219
Getter surface [cm ²]	1,076
Flange size	CF100 (rotatable)
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	6.5
Maximum bake-out temperature [°C]	250*
ION element type	Noble diode

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

Activation and working parameters

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

The pump is equipped with a type-K (Chromel-Alumel) thermocouple.

Type of operation	Temperature*	Electrical parameters
NEG activation	510 °C	195 W (46.5 V) [†]
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity ‡ [Torr·l]
	Activated NEG [§]	Saturated NEG ^{**}	
H ₂	1,150	13	3,650 ^{††}
H ₂ O ^{‡‡}	850	9	400
CO	500	11	7
N ₂	320	9	4
CH ₄	32		50,000 h at 10 ⁻⁶ Torr
Ar	10 (2.5) ^{§§}		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

* Thermocouple reading when the getter material is at 550 °C (activation).

† Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

‡ Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

§ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

** Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

†† When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

‡‡ Calculated values.

§§ Measured at 3·10⁻⁶ Torr. Unsaturated ION element (saturated ION element).

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive, *i.e.*, a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N_2 , air, or Ar) to vent the pump, avoiding the use of atmospheric air.

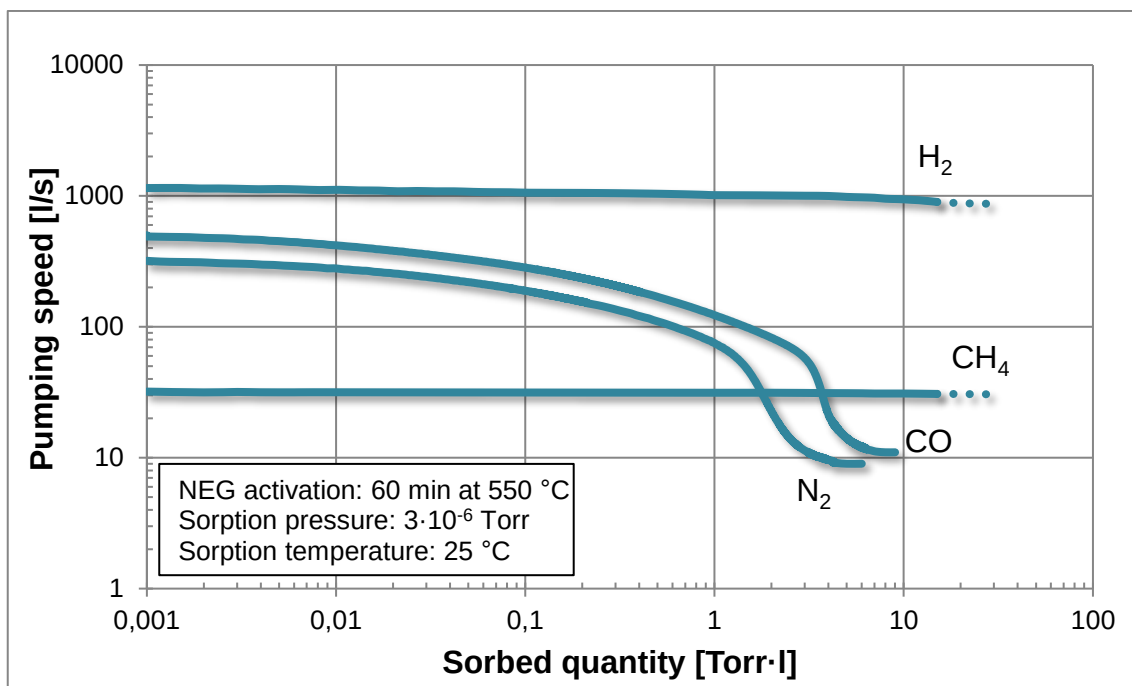


Figure 14 – Typical sorption curves of NEXTorr Z 1000 (according to ASTM F798-97).

ION element pumping characteristics

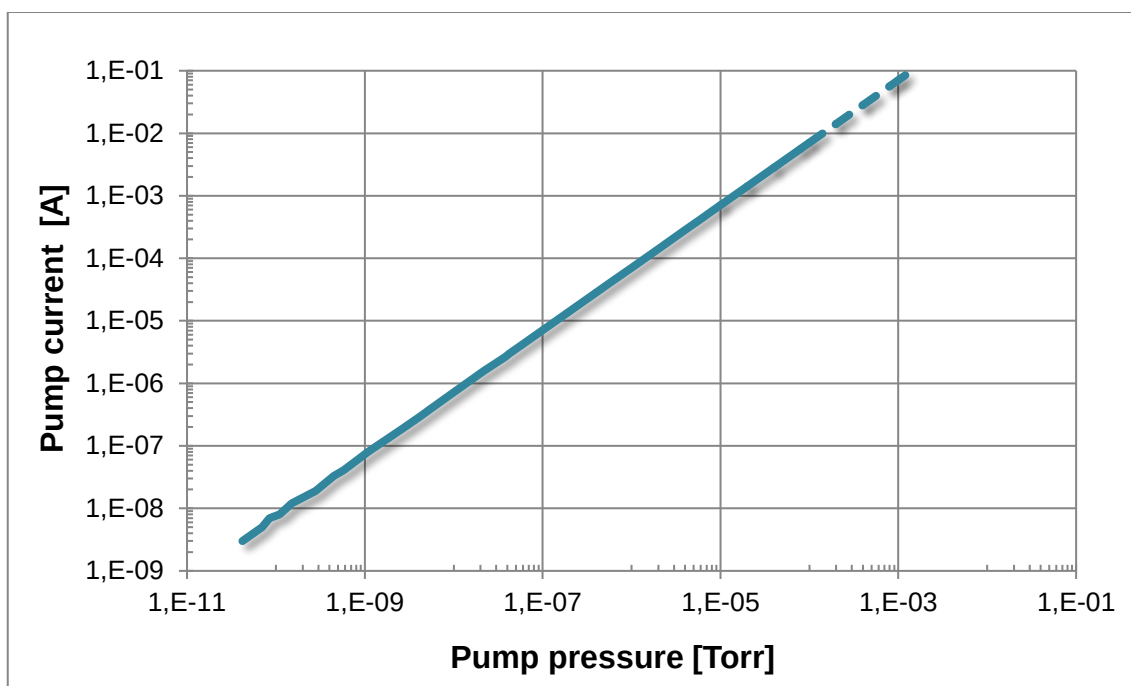


Figure 15 – ION current vs. pressure characteristic curve for N₂ of NEXTorr Z 1000.

Ordering information

Product	Product description	Code
NEXTorr pump	NEXTorr Z 1000	5H0237
NEG element power supply	NEG POWER C1*	3B0501
NEG element output cable	NEG cable 6P5A 3 MT†	3B0854
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

NEXTorr Z 2000

Pump dimensions

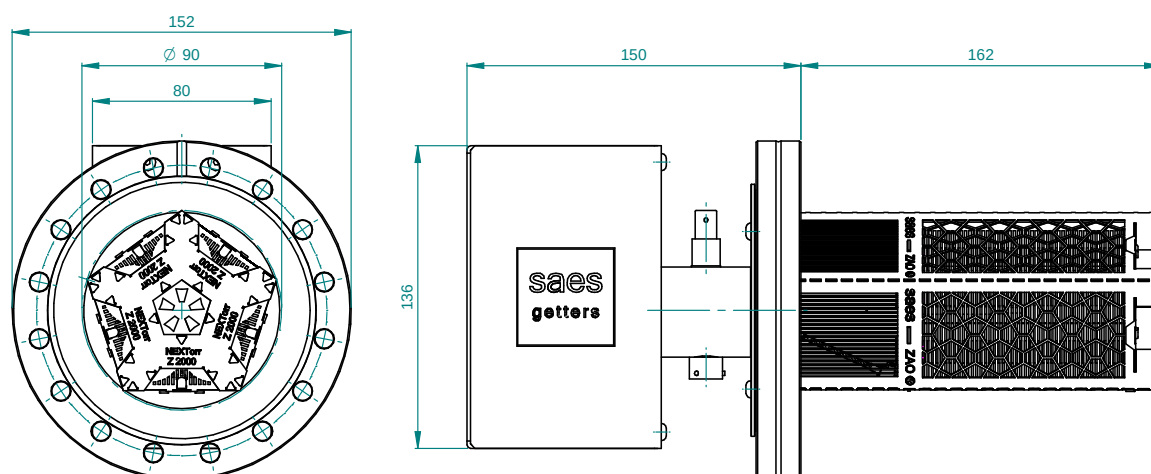


Figure 16 – 2D drawings of NEXTorr Z 2000. All dimensions are in mm. Indicated dimensions may change and are for reference only; refer always to DWG M.H220D.0006.22.

Main technical features

Getter alloy	ZAO® 1
Alloy composition	Ti Zr V Al
Getter mass [g]	430
Getter surface [cm ²]	2,350
Flange size	CF100 (rotatable)
Flange material	AISI 304
Cage material	AISI 316L
Total weight incl. magnets [kg]	6.7
Maximum bake-out temperature [°C]	250*
ION element type	Noble diode

Activation and working parameters

* This bake-out temperature is valid for the pump body only. If magnets and red covers are not removed, the maximum bake-out temperature is **150 °C**.

To start the NEG activation process, the system has to be pumped down to a pressure of $1 \cdot 10^{-4}$ Torr or lower. Typically, the activation should last **1 h**.

Once activated, no power is needed to operate the NEG element at RT in UHV conditions. The maximum current to the NEG element is **5 A**. The maximum ION element operating voltage is **5 kV dc**.

The pump is equipped with a type-K (Chromel-Alumel) thermocouple.

Type of operation	Temperature*	Electrical parameters
NEG activation	510 °C	230 W (64 V) [†]
NEG working (UHV)	RT	No power needed

Pumping speed and capacity

Gas	Pumping speed [l/s]		Sorption capacity ‡ [Torr·l]
	Activated NEG [§]	Saturated NEG ^{**}	
H ₂	2,200	13	6,840 ^{††}
H ₂ O ^{‡‡}	1,600	9	700
CO	920	11	11
N ₂	640	9	6
CH ₄	32		50,000 h at 10^{-6} Torr
Ar	10 (2.5) ^{§§}		---

Working conditions	Reactivations (sorption cycles)
RT (UHV)	>100

The total capacity for each gas is reached after many reactivations, when the getter is fully consumed. Capacity values are not additive,

* Thermocouple reading when the getter material is at 550 °C (activation).

† Default activation parameters, assuming that the NEG cartridge is not confined in a tight tube or volume.

‡ Single-run capacity is reached when pumping speed is equal to the pumping speed of the ION element only. This limit does not apply for H₂.

§ Initial pumping speed of a fully-activated pump, without significant conductance limitations.

** Pumping speed of the ION element only (saturated NEG), without significant conductance limitations.

†† When the getter reaches its H₂ capacity, it can be *regenerated*. Through the regeneration process, it is possible to extract the H₂ content stored inside the bulk. Refer to the appendix at the end of this document for further information.

‡‡ Calculated values.

§§ Measured at $3 \cdot 10^{-6}$ Torr. Unsaturated ION element (saturated ION element).

i.e., a getter fully reacted with one gas species will not sorb any other gas.

To maximise the number of possible reactivations, it is strongly recommended to use a dry gas (N₂, air, or Ar) to vent the pump, avoiding the use of atmospheric air.

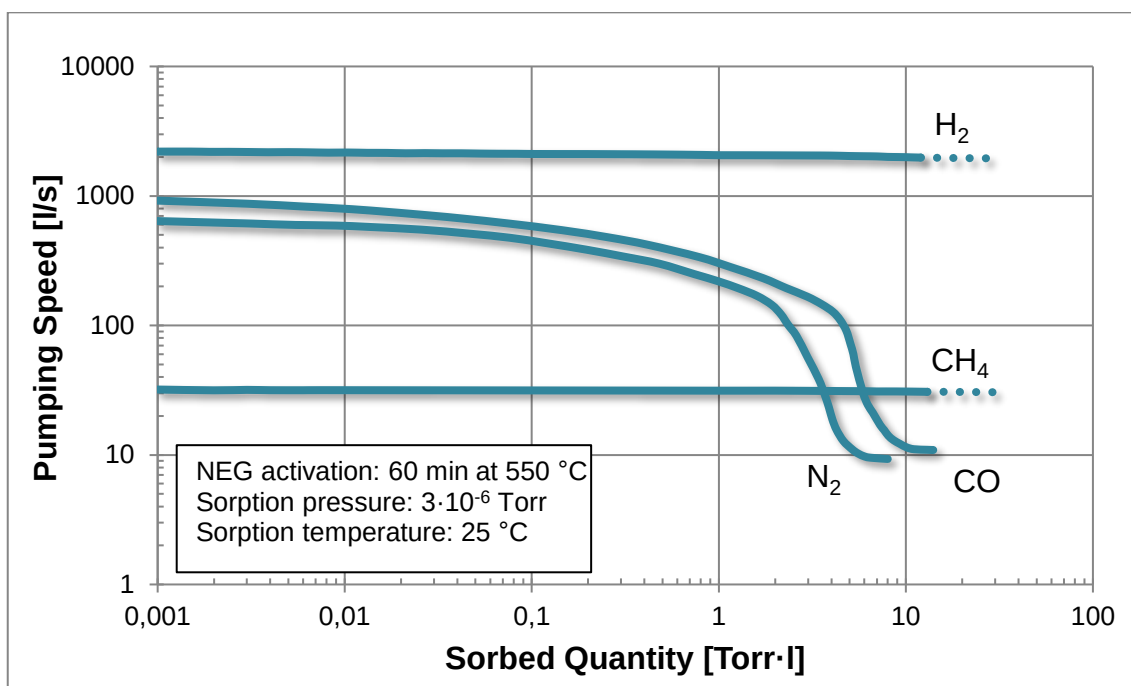


Figure 17 – Typical sorption curves of NEXTorr Z 2000 (according to ASTM F798-97).

ION element pumping characteristics

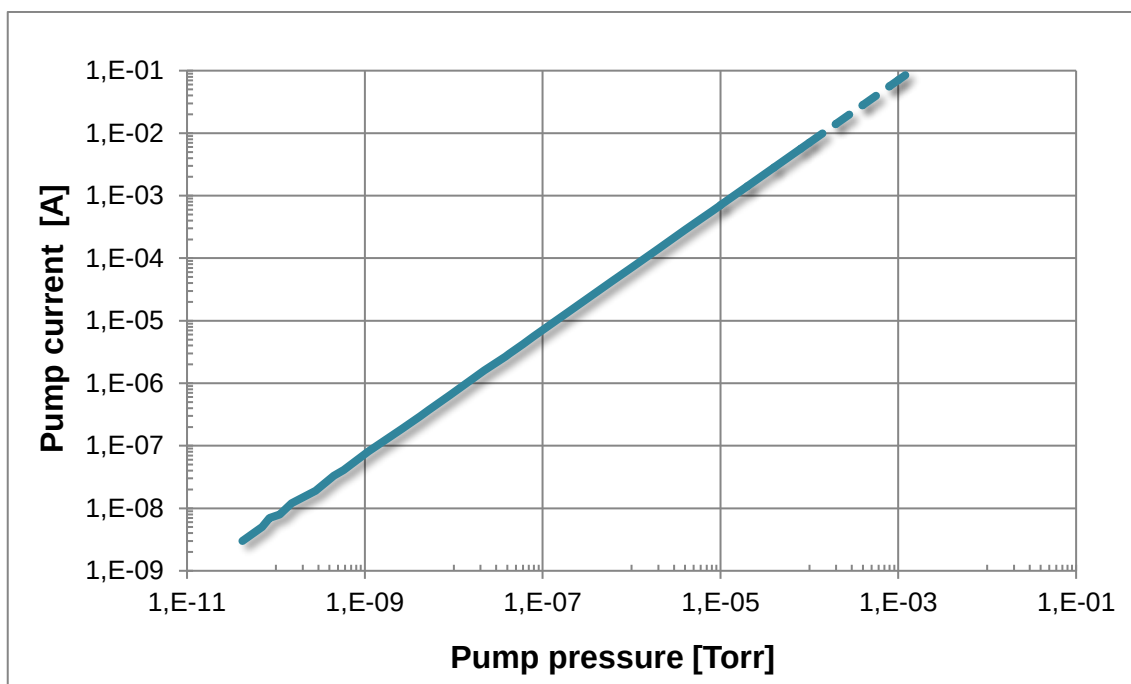


Figure 18 – ION current vs. pressure characteristic curve for N₂ of NEXTorr Z 2000.

Ordering information

Product	Product description	Code
NEXTorr pump	NEXTorr Z 2000	5H0244
NEG element power supply	NEG POWER C1*	3B0501
NEG element output cable	NEG cable 6P5A 3 MT†	3B0854
ION element power supply	SIP POWER‡	3B0506
ION element power supply	SIP POWER LEDS RS485§	3B0509
ION element output cable	IP CABLE**	3B0568

* Other models, able to simultaneously drive up to four pumps, are available.

† Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

‡ With display.

§ Without display.

** Cable is 3 m long, radiation resistant (10⁹ rad) and bakeable up to 250 °C. Longer cables are available on request.

Appendix: NEG regeneration

Unlike the other getterable gases, H_2 does not form stable chemical bonds with the getter material: the molecule dissociates at the getter surface and hydrogen atoms diffuse into the bulk material, driven by a concentration gradient.

This means that the pumping of hydrogen and its isotopes by a NEG pump is a reversible process: a getter will either sorb or release H_2 to meet the equilibrium pressure, which depends on the concentration of hydrogen atoms dissolved into the bulk, according to Sieverts' law.

The so-called *embrittlement* limit is reached when the quantity of hydrogen inside the getter is high enough to modify the mechanical characteristics of the alloy, which can ultimately lead to the getter flaking off in small particles, due to the formation of brittle metal hydrides, the creation of voids at the grain boundaries and the propagation of cracks.

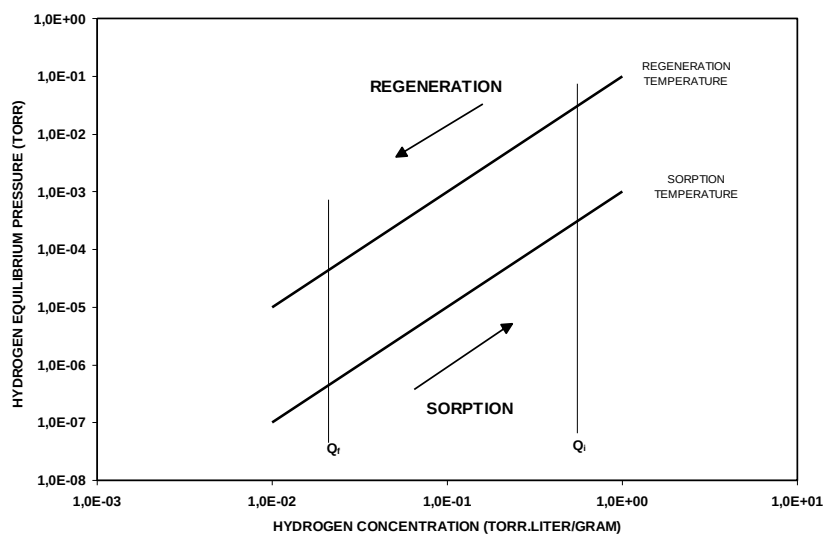


Figure 19 – Regeneration mechanism of a getter pump.

Hydrogen and its isotopes can thus be easily extracted from the getter material through a *regeneration* treatment, which becomes necessary when:

- H_2 pumping speed falls below a minimum acceptable limit, because the equilibrium pressure is approaching;

- equilibrium is still far but the quantity of hydrogen or hydrogen isotopes stored inside the bulk is close to the embrittlement limit.

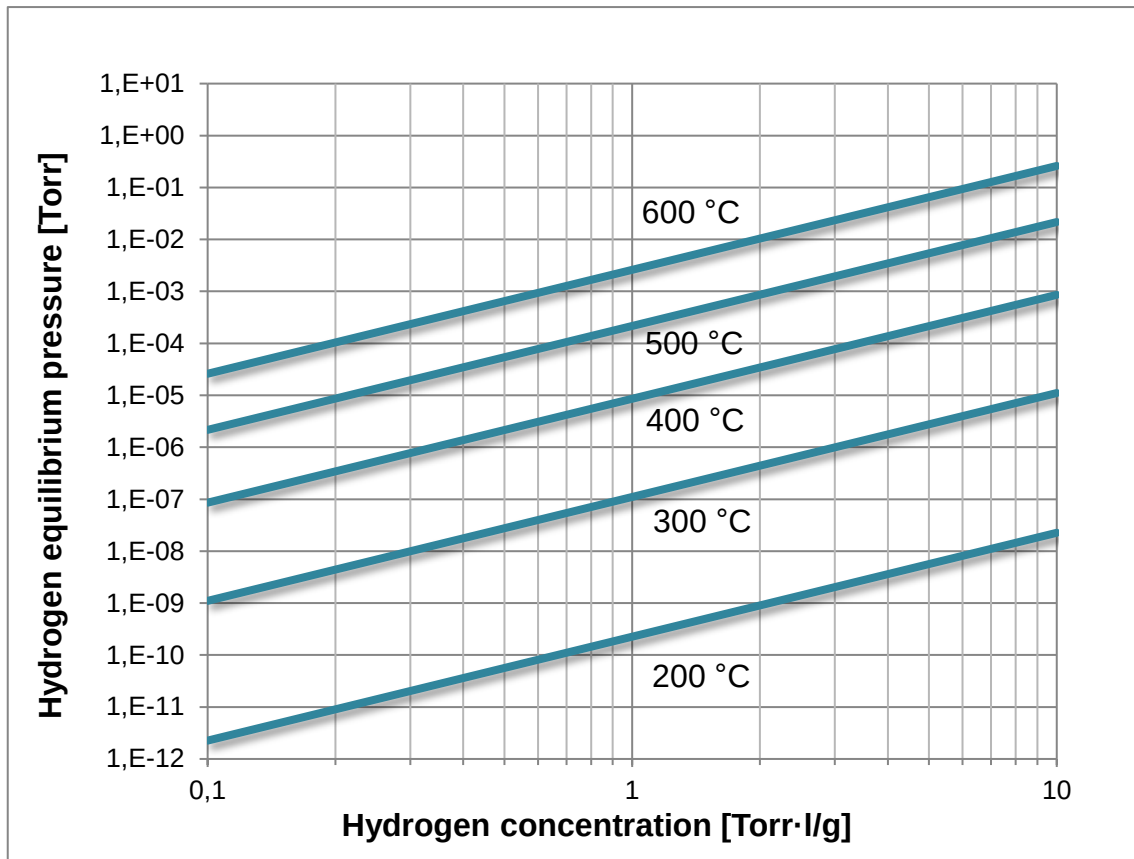


Figure 20 – ZAO hydrogen equilibrium isotherms.

By increasing the temperature of the getter material, its hydrogen equilibrium pressure also increases (cf. Figure 19); hence, by means of a standard backing pump, it is possible to remove the H₂ molecules released by the getter. The higher is the temperature, the more efficient the regeneration process will be.

The time necessary to regenerate a NEG pump is given by the following expression:

$$t = \frac{M}{F} \cdot \left(\frac{1}{q_f} - \frac{1}{q_i} \right) \cdot 10^{-(A-B/T)}$$

where:

- t [s] is the regeneration time;
- M [g] is the mass of getter material;
- F [l/s] is the pumping speed of the backing pump;
- q_f [Torr·l/g] is the final hydrogen concentration;
- q_i [Torr·l/g] is the initial hydrogen concentration;
- A is equal to 5.77 for ZAO alloy;

- B is equal to 7290 for ZAO alloy;
- T [K] is the regeneration temperature.

Since the regeneration process follows an exponential law, a significant amount of time can be saved if a partial regeneration is achieved (e.g., 90% of the stored hydrogen is extracted instead of 100%). For the same reason, if a NEG pump must sorb a given amount of hydrogen during each cycle (e.g., 5 Torr·l/g), it is more convenient to operate in the $q_f=7-q_i=2$ region than in the $q_f=5-q_i=0$ one.

ZAO hydrogen equilibrium isotherms are given in Figure 20.



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